

Raise Your Eyebrows: Investigating the Impact of Virtual Avatar’s Facial Expression Scaling on Users’ Perception of Emotion and Uncanny Valley Effect in Social Virtual Reality

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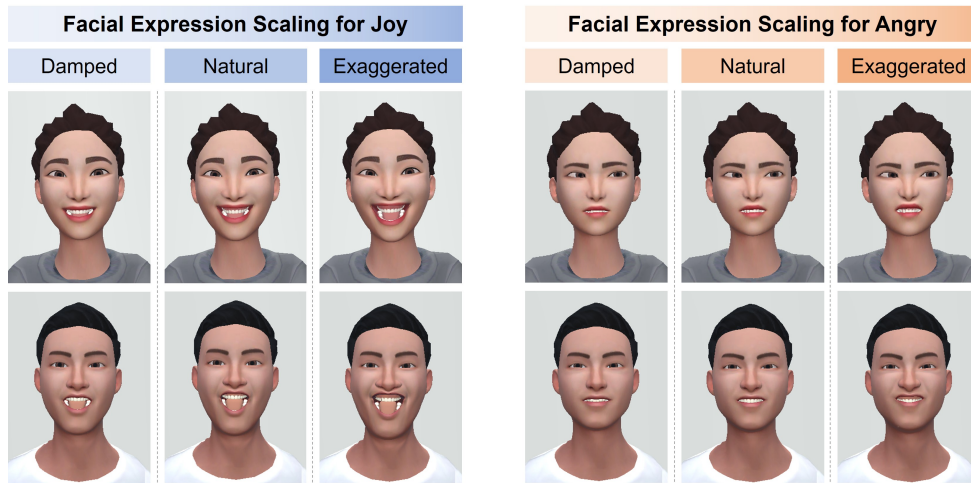


Fig. 1. Illustrations of facial expression scaling on virtual avatars for Joy and Anger, demonstrating three levels: Damped (reduced amplitude), Natural (accurately captured amplitude), and Exaggerated (enhanced amplitude). Left: Joy expressions are naturally pronounced, particularly in the mouth region, and can appear excessively intense when exaggerated. Right: Anger expressions, where upper facial features (eyebrows, eyelids, and cheeks) often display minimal movement and are barely perceptible in their natural state, become more vivid and recognizable when exaggerated. These representative moments highlight how exaggeration can either diminish or enhance the perceived realism of the expression depending on the emotion portrayed.

Facial expression scaling, which artificially exaggerates or damps the amplitude of expression, has been shown to enhance expressiveness but risks triggering the uncanny valley effect. This study explores the effects of facial expression scaling on cartoon avatars in social VR using a mixed-methods approach with 30 participants. We assessed users’ emotional perception and uncanny valley responses while they viewed scaled, pre-recorded facial expressions in simulated face-to-face VR interactions. Results revealed that although expression exaggeration did not significantly influence the accuracy of emotion perception, it led to significant interaction effects with emotion categories on uncanny valley effect. Notably, exaggerated expressions both diminish or enhance the perceived realism, eeriness, and appeal of the expression, depending on the specific emotion conveyed. Joy and Surprise expressions, particularly in the mouth region, are naturally pronounced and can appear unnatural when exaggerated further. Conversely, upper facial expressions for Fear, Sadness, Disgust, and Anger—involving eyebrows, eyelids, and cheeks—are often subtle and hard to perceive, and exaggeration enhances their vividness and realism. Based on these findings, we developed a selective facial expression exaggeration strategy focusing on upper facial features. A validation study with 20 participants confirmed the effectiveness of this approach.

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CCS Concepts: • **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Virtual reality**.

Additional Key Words and Phrases: Social Virtual Reality, Facial Expression Scaling, Cartoon Avatars, Emotion Perception, Uncanny Valley Effect, Mixed-Methods Study

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1 INTRODUCTION

Social Virtual Reality (VR) has gained prominence as a leading application within virtual environments [62]. Using immersive head-mounted displays (HMDs), users from disparate locations can engage in social interactions within a shared virtual space, simulating physical co-presence [26]. These environments allow users to craft digital self-representations, or avatars, facilitating interactions that closely emulate face-to-face communication [70]. Unlike other avatar-based systems (e.g., Memoji, MMORPGs, and 3D virtual worlds) that primarily support on-screen avatar-mediated communication, full-body tracked avatars in social VR enable a broader array of communication modes, including both verbal and non-verbal interactions such as voice, hand gestures, body postures, gaze, and facial expressions [67, 88]. In addition, the ability to express and perceive emotions in these virtual environments is crucial for building meaningful social connections. Emotional expression and recognition facilitate empathy, trust, and intimacy, which are fundamental components of human social interaction and relationship formation in both physical and virtual spaces [65, 92].

In social-emotional communication, facial expressions are crucial components of non-verbal behavior, conveying the emotional state or intention of the expressor [25, 47, 99]. Recent studies highlight the importance of realistic facial expressions in virtual avatars for enhancing users’ sense of face-ownership [28, 55] and for effectively conveying emotions [3, 31]. Consequently, the latest generations of HMDs, such as Meta Quest Pro, Apple Vision Pro, and Pico 4 Pro, are equipped with inward-facing cameras for real-time facial tracking. A common approach to achieve natural facial expressions on virtual avatars involves translating the facial movements detected by the headset’s sensors into the activation of expressions (e.g., jaw drop, nose wrinkle) as defined by the Facial Action Coding System (FACS) [22]. However, human facial expressions are highly nuanced and diverse [16]. Due to the limited model granularity of cartoon avatars and the simplifications inherent in the FACS for facial muscle movements [72], the authenticity and naturalness of cartoon avatar facial expressions are significantly compromised [28, 56, 64]. Virtual facial expressions are often rated lower on intensity, arousal, and valence than human ones [51, 71].

Previous research has demonstrated that exaggerating facial expressions of static cartoon avatars can enhance emotional intensity and improve avatar-mediated communication experiences [30, 81]. Moreover, research indicates that avatars with real-time tracked facial expressions and amplified smile coefficients evoke greater positive affect and enhance social presence compared to those with unmodified smiles [76]. Conversely, research suggests that dampened facial motion can enhance the likability of cartoon avatars [44]. However, modifying facial expression amplitude carries potential risks, including the uncanny valley effect, which can cause user discomfort, potentially disrupting communication flow, reducing empathy, and diminishing the overall sense of presence and connection between users in social VR interactions [45, 66].

Investigating the impact of facial expression scaling on real-time HMD-tracked expressions could significantly inform the design of virtual avatars in social VR. This research has the potential to enhance users' perception of emotions on virtual avatars without triggering the uncanny valley effect. However, there is a notable gap in the literature regarding how such scaling affects user experience and the communication of various emotion types in social VR contexts. To address this, our study aims to answer the following research questions:

- **RQ1:** How does facial expression scaling influence users' perception of others' emotions in social VR?
- **RQ2:** What is the relationship between facial expression scaling and the uncanny valley effect across different emotional expressions?
- **RQ3:** How can we leverage facial expression scaling to enhance emotional communication in social VR while mitigating potential uncanny valley effects?

To address these research questions, we conducted two mixed-method studies using carefully prepared materials. First, we collected dynamic facial expressions representing six basic emotions [23] (joy, sadness, fear, anger, surprise, and disgust) from trained actors during natural speech scenarios. These real-time expressions, captured by HMD facial tracking, were then mapped onto cartoon avatars and scaled at three levels: damped, natural, and exaggerated, as illustrated in Figure 1.

In Study 1, 30 participants engaged in multi-modal interactions within a simulated face-to-face VR setting, both viewing the pre-recorded, scaled dynamic expressions and listening to the corresponding speech. They assessed emotion perception and uncanny valley effects across these audio-visual stimuli, using both scales and open-ended responses. The independent variables were the facial expression scaling level and emotion category, while dependent variables included measures of emotion recognition accuracy, perceived emotional intensity, emotional contagion, and various aspects of the uncanny valley effect (including perceived facial vividness, realism, appeal, consistency with speech, and comfort). Quantitative analysis revealed that expression exaggeration did not significantly influence participants' emotion perception accuracy but led to decreased facial realism and comfort, and increased eeriness. We observed a significant interaction effect between expression scaling and emotional categories on uncanny valley indicators. For joy, surprise, and disgust, exaggerated expressions increased facial eeriness and decreased appeal, realism, and comfort. Conversely, for fear, sadness, and anger, facial exaggeration enhanced realism and appeal.

Analysis of subjective feedback showed that users rely on multiple cues, including speech and context, for emotion recognition, demonstrating the robustness of emotional communication in social VR despite manipulated visual expressions. The interaction effect on the uncanny valley stems from inherent differences in expression magnitudes across emotions. Joy, surprise, and disgust expressions, with naturally pronounced mouth and cheek movements, become visually discomforting when exaggerated further. However, for fear, sadness, and anger, characteristic facial expressions (e.g., eyebrow furrowing, nose wrinkling) are typically subtle and stiff, possibly due to HMD pressure on the forehead limiting upper facial movements. Exaggeration makes these subtle movements more recognizable, enhancing perceived naturalness and realism.

Based on these findings, we developed a selective facial expression exaggeration strategy: highly exaggerate eyebrows, moderately exaggerate eyelids and nose, weakly exaggerate cheeks, and avoid exaggeration in lower face regions. A validation study with 20 participants, using pairwise comparisons, assessed this strategy against unmodified expressions. Results showed that our approach significantly enhanced users' perceived emotional intensity and contagion and reduced the uncanny valley effect for fear, disgust, and anger, while maintaining neutral or slightly positive impacts for joy, surprise, and sadness.

Our contributions can be summarized as follows:

- Our study reveals that facial expression scaling in cartoon avatars doesn't significantly affect emotion perception accuracy in multi-modal social VR interactions, indicating users rely on multiple cues for emotion recognition.
- We identified that facial expression scaling interacts significantly with emotion categories on the uncanny valley effect. Exaggerated expressions mitigate this effect for Fear, Sadness, Anger by enhancing subtle upper facial cues, while potentially exacerbating it for Joy, Surprise, and Disgust due to over-emphasis of already pronounced mouth features.
- We proposed and validated a selective facial expression exaggeration strategy that effectively reduces the uncanny valley effect for disgust, fear, and anger while maintaining or slightly improving it for joy, surprise, and sadness, offering practical guidelines for enhancing cartoon avatars in social VR.

2 RELATED WORKS

In this section, we review current research on avatar facial expressions in social VR and highlight how our work addresses existing gaps. We begin by examining avatar facial enhancement strategies and their impact on users' emotional perceptions of virtual avatars. Next, we explore the factors influencing emotion perception through facial expressions in social VR environments. Finally, we discuss previous studies investigating how avatar faces in social VR can trigger the uncanny valley effect, and how our research contributes to this ongoing discourse.

2.1 Facial Enhancement Strategies and Their Effects on Users' Emotional Perceptions of the Virtual Avatar

Facial expressions are crucial for conveying and interpreting emotional states, thereby facilitating effective communication in social interactions [82]. In social VR, various enhancement approaches have been developed to improve avatar facial expressiveness, including real-time facial expression tracking and techniques to make expressions appear more realistic and natural.

Tracking facial expressions in XR environments presents unique challenges due to the obstruction of large facial areas by head-mounted displays. Prior to the advent of headsets with built-in facial tracking, researchers employed custom hardware or integrated existing sensors, often combining lower face tracking with eye-tracking capabilities [61, 78, 85]. Studies have explored how dynamic facial expressions influence user interactions in virtual environments. Kokkinara et al. demonstrated that synchronous mapping of facial tracking onto virtual faces can induce high levels of face-ownership and agency [55]. Hartbrich et al. further confirmed the benefits of integrated eye and face tracking for certain enfacement scales [32].

Research has also shown that increasing animation realism through real-time tracking enhances the perceived appeal of virtual faces [55]. Moreover, dynamic facial expressions, particularly eye and mouth movements, have been found to increase social presence in collaborative virtual tasks [52]. However, accurately replicating the nuanced details of human facial expressions in avatars remains a significant challenge due to their complexity and subtlety. Gaps in behavioral realism of avatar facial expressions can affect users' emotional perception accuracy, potentially impacting the quality of communication in virtual environments [34]. Moreover, the authenticity and fidelity of facial appearance play crucial roles in facilitating honest social communication in virtual environments. Research by Dubosc et al. examined how facial anthropomorphism impacts body ownership, avatar attractiveness, social presence, and performance [18]. Similarly, Kang et al. investigated the effects of virtual character face detail levels on social presence and interpersonal communication in augmented reality collaborations [49].

For cartoon-style avatars, exaggerated facial movements are widely employed to enhance emotional cues [44]. Early work by Hart et al. suggested that augmenting facial expressions could improve social VR experiences [30]. Oh et al. further demonstrated that enhanced smiles led to more positive affect and stronger social presence compared to normal smiles [76].

Facial expression scaling, which modulates the intensity of facial movements, addresses the balance between animation realism and emotional authenticity in cartoon avatars. This technique aims to enhance emotional genuineness while preserving stylized aesthetics, potentially improving both communication fidelity and user experience in social VR. However, a significant research gap exists regarding the effects of such scaling on user experience and emotion communication in social VR. While studies have examined facial motion’s impact on personality perceptions [46] and positive affect [76], the influence of systematically scaled expressions on emotion recognition, perceived authenticity, and the uncanny valley effect remains unexplored, especially for real-time tracked faces in immersive environments.

2.2 Perception of Avatar’s Emotion

Emotion perception in virtual environments relies on multiple channels, mirroring real-world interactions. Schirmer and Adolphs [89] highlight how signals from face, voice, and touch converge to form a holistic understanding of emotions. In virtual contexts, facial expressions play a fundamental role in non-verbal communication [21, 24], serving as a primary channel for emotional inference.

Research has demonstrated that humans can accurately recognize facial expressions of both humanoid and non-humanoid virtual agents, comparable to real human expressions [20, 33, 39]. Factors influencing emotion recognition in virtual environments include interpersonal distance [13], visual realism [36], expression driving methods [40, 41, 97], avatar anthropomorphism [93], personalization [84], and character type [4]. Additional visual cues, such as biosignal visualizations, have been shown to enhance emotional perception and empathy towards avatars [60, 101].

Voice plays a crucial role in emotion recognition within virtual environments [59, 91], conveying information beyond semantics [2]. This aspect is particularly relevant to our research, as we investigate the interplay between facial expressions and voice in emotion perception. Some studies have explored user perceptions when emotions conveyed through different channels are inconsistent [10, 102].

However, a significant gap exists in understanding how emotions conveyed through face-tracked avatar expressions align with those communicated through voice and posture in virtual environments. Our research addresses this gap by examining the consistency and interaction between facial expression scaling and vocal cues in emotion perception. By investigating how scaled facial expressions interact with vocal information, we provide insights into the complex multimodal nature of emotion communication in social VR, contributing to a more comprehensive understanding of virtual emotional interactions.

2.3 Uncanny Valley Effect

The uncanny valley effect, first proposed by Mori in 1970 [74], describes a phenomenon where affinity towards humanoid figures initially increases with human likeness, then sharply decreases at a certain point, before rising again as the figure becomes nearly indistinguishable from a real human [69]. This effect has significant implications for social interactions in virtual environments, potentially disrupting user engagement and emotional connection [90].

In VR contexts, avatar realism has been identified as a key factor contributing to the uncanny valley effect, with highly realistic avatars potentially eliciting stronger negative reactions [57, 64]. Conversely, cartoon-like characters often

prove more attractive and manageable in terms of facial expressions. The uncanny valley phenomenon is influenced by both avatar appearance and behavior [7, 50], with the upper face region being particularly critical [95, 103].

Enhancing facial expressions presents a complex challenge: while it can intensify emotional conveyance, it may also trigger the uncanny valley effect. Hyde et al. [45] found that enhanced facial expressions positively correlated with perceived extroversion and persuasiveness but negatively impacted perceived realism and naturalness. Mäkäräinen et al. [66] suggested that high exaggeration is suitable only for less realistic avatars, as increased realism can easily lead to the uncanny valley effect. Furthermore, the intensity of emotions expressed by virtual humans influences uncanny valley ratings [36].

Our research addresses a significant gap in the literature by investigating how the amplification of dynamic expressions, tracked by built-in VR headset cameras, affects the uncanny valley effect. We explore the nuanced relationship between emotion types, expression scaling, and the uncanny valley phenomenon in social VR contexts. This approach provides insights into optimizing facial expression enhancement for different emotions while mitigating potential negative effects, contributing to the development of more effective and engaging virtual social interactions.

3 MATERIALS

To analyze the impact of facial expression scaling on virtual characters while minimizing confounding variables, we developed a comprehensive material collection and production process. Our approach involved recording, scaling, and replaying facial expressions, inspired by Kullmann et al. [56]. This method allows for controlled manipulation of facial expressions while maintaining consistent content across conditions, crucial for isolating the effects of expression scaling.

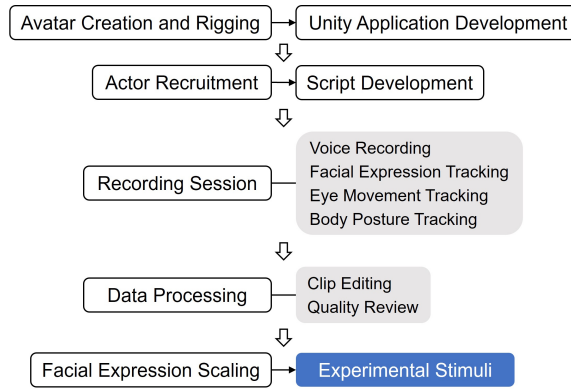


Fig. 2. Flowchart illustrating the preparation process for dynamic facial expression materials in our study.

As shown in the process diagram (Figure 2), our material preparation involved several interconnected stages. We began with avatar creation and Unity application development, concurrent with actor recruitment and script development. Trained actors then performed emotional expressions in a simulated VR environment, where we recorded their voice, facial expressions, eye movements, and body postures using HMDs. Following data processing, we meticulously edited all recorded clips, trimming irrelevant sections such as blank periods and greetings. We then thoroughly reviewed each clip to ensure data quality and accuracy, maintaining a comprehensive set of expressions for each emotion category.

These expressions were then scaled to create our experimental stimuli. The following sections detail each stage of this process, including the material preparation and data collection procedures.

3.1 Apparatus and Avatar Creation

To collect and replay actors' facial expressions, voice, and movements during emotional expression, we employed a custom Unity 2021.3.26f1 application running on ASUS laptops and streamed to Meta Quest Pro headsets. The Meta Quest Pro features a 72Hz refresh rate, 1800×1920 resolution per eye, and a 106° horizontal by 96° vertical field of view. Our laptops ran Windows 11 with Intel i9-12900H CPUs, 32.0 GB of memory, and NVIDIA GeForce RTX 3080 Ti GPUs.

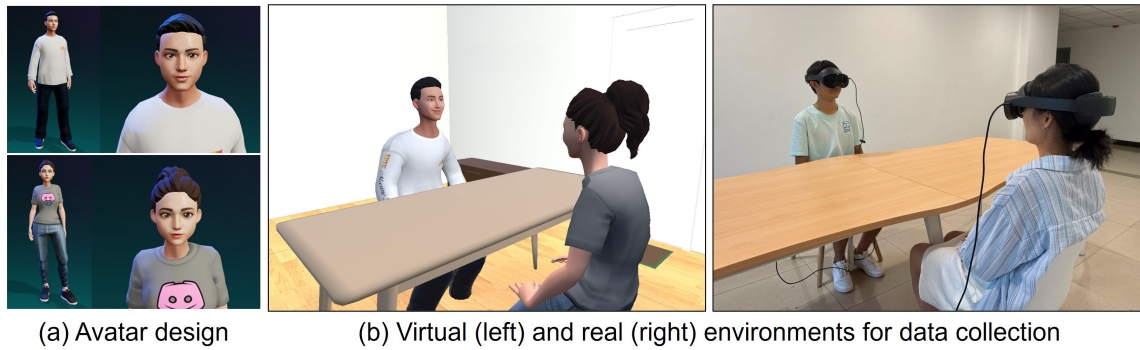


Fig. 3. An illustration of the cartoon avatar used in our study (a) and the virtual and real environments for collecting emotional expression data (b).

We utilized cartoon-style avatars, aligning with current social VR platforms like Meta Horizon Worlds, VR Chat, Roblox, and Rec Room. This choice ensures sufficient embodiment, visual appeal [64], reduced uncanny valley effects [58], greater customizability [1], and privacy protection [63], enhancing the generalizability of our findings to most social VR applications. We created full-body avatar models using Ready Player Me¹, a cross-game platform used by major social VR applications. Full-body avatars were chosen based on research showing their superior performance over partial body configurations [99]. As shown in Figure 3(a), we created one male and one female generic avatar. A professional 3D animation artist added 70 facial blendshapes to each avatar, following Face Tracking API guidelines. We then implemented scripts in Unity to map detected expressions to these avatar blendshapes.

Our application used the Oculus Movement SDK for Unity for body, face, and eye tracking. The Face Tracking API² uses inward-facing cameras to detect facial movements, categorizing them based on the Facial Action Coding System (FACS) [22]. Each expression is represented as a blendshape with a strength ranging from 0% (completely relaxed) to 100% (maximum amplitude), indicating the intensity of the facial expression.

3.2 Actor Recruitment and Script Development

To ensure natural and realistic performances while minimizing individual expression biases, we recruited five pairs of trained actors, each pair comprising one male and one female. The actors' demographics were matched with the

¹<https://readyplayer.me>

²<https://developer.oculus.com/documentation/unity/move-face-tracking>

avatars' appearances to enhance realism. Our cohort consisted of 10 Chinese actors from a university drama team, averaging 22 years of age ($SD = 1.9$), with 2-3 years of training and stage experience.

Actors were asked to perform according to simple scripts in a simulated conversational scenario, a method validated by Kullmann et al. [56]. The actors were instructed to narrate a personal experience to a friend (the experimenter) while conveying a designated emotion. We focused on Ekman's six basic emotions [23]: Joy, Sadness, Fear, Anger, Surprise, and Disgust, chosen for their universal recognition across cultures.

For each emotion, we prepared three distinct scripts, resulting in a total of 18 scripts. All scripts were written in Chinese to match the native language of our actors and participants. Each script comprised an emotional monologue designed for approximately 30 seconds of delivery, a duration chosen to capture a full emotional arc while maintaining participant engagement. We randomly distributed the 18 scripts between each pair of actors, with each actor performing nine scripts.

Script content was tailored to match both the intended emotions and the life experiences of university students, as our actors were primarily from this demographic. This approach ensured relatable and authentic performances. Following Yang et al. [102], one author generated initial scripts based on common conversation topics shared in online videos. The entire research team then collaboratively reviewed, discussed, and refined these scripts to meet our criteria: appropriate length, natural language use, and accurate emotional conveyance. To ensure transparency and reproducibility, we provide detailed translated versions of all scripts in the Supplemental Material. This allows other researchers to understand the specific stimuli used and potentially replicate or build upon our work.

3.3 Emotional Expression Collection

To facilitate motion data collection via HMDs while maintaining authentic performances, we created a virtual environment simulating face-to-face conversations (Figure 3(b)). This setup mirrored the real-world arrangement where the experimenter and actor sat across a table. In the virtual space, gender-matched avatars represented both parties, enabling real-time visualization of facial expressions and body movements, along with voice communication, similar to platforms like VRChat.

Prior to data collection, we carefully calibrated the HMDs to prevent posture abnormalities. The actors, having undergone VR-specific emotional expression training, were given 3-5 minutes to prepare each script. The recording sessions began with friendly greetings, followed by the actor's emotional narration of the scripted experience. The experimenter provided non-verbal feedback (e.g., nodding, smiling) to maintain conversational realism. We recorded the actors' voices, facial expressions, gaze directions, and body postures throughout. Post-performance, actors rated their emotional arousal on a 7-point Likert scale. A 3-minute interval between performances allowed actors to reset emotionally, ensuring distinct emotional states for each recording.

We collected 90 emotional expression clips from the five actor pairs, averaging 32.2 seconds in duration ($SD = 8.3$). These clips underwent meticulous editing to remove irrelevant sections such as blank periods and greetings, focusing on the core emotional content. We conducted thorough reviews to ensure data quality and accuracy, maintaining the integrity of each emotional expression. This comprehensive approach to data collection and preparation allowed us to create a robust dataset of natural, diverse emotional expressions, crucial for the validity of our subsequent analyses.

3.4 Overview of Collected Expressions

Table 1 presents the average activation levels of facial blendshapes across various emotional expressions, with symmetric blendshapes averaged. The data reveal significant variations in blendshape activation across facial regions and emotions.

Notably, *LidTightener*, *CheekRaiser*, *UpperLipRaiser*, *LipCornerPuller*, *Dimpler*, and *JawDrop* showed the highest activation levels.

Table 1. The average strength of different facial blendshapes across various emotional expressions, with darker colors indicating higher strength. Symmetric facial blendshapes were averaged.

Facial region	Blendshapes	Joy	Surprise	Disgust	Fear	Sadness	Anger
Eyebrows	BrowLowerer	5.18%	3.28%	7.09%	5.30%	3.93%	5.59%
	InnerBrowRaiser	2.93%	2.11%	2.42%	1.82%	0.91%	1.38%
	OuterBrowRaiser	1.57%	1.76%	1.60%	1.34%	0.45%	1.02%
Gaze	EyesClosed	8.97%	7.90%	11.10%	8.14%	9.62%	9.36%
	EyesLookRight	7.74%	8.56%	7.33%	6.60%	9.40%	7.96%
	EyesLookLeft	2.17%	3.33%	3.83%	7.79%	3.11%	3.27%
	EyesLookDown	4.35%	2.27%	4.09%	2.52%	5.45%	3.27%
	EyesLookUp	0.88%	2.86%	0.95%	1.96%	0.11%	0.72%
Eyelid	LidTightener	21.97%	11.86%	16.73%	8.63%	3.42%	10.49%
	UpperLidRaiser	0.18%	0.46%	0.93%	0.99%	0.20%	0.69%
Cheeks	CheekRaiser	26.79%	12.11%	11.89%	5.95%	4.89%	8.05%
	CheekPuff	0.04%	0.04%	0.07%	0.06%	0.05%	0.04%
	CheekSuck	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Nose	NoseWrinkler	1.28%	0.82%	5.05%	1.96%	0.56%	2.12%
Mouth	UpperLipRaiser	38.05%	25.47%	24.22%	13.01%	9.16%	16.66%
	LipCornerPuller	36.45%	16.80%	9.49%	3.19%	1.69%	4.77%
	Dimpler	20.54%	12.16%	7.82%	4.04%	3.14%	5.73%
	ChinRaiser	9.17%	6.38%	6.89%	3.62%	2.82%	4.77%
	LipSuck	9.17%	5.20%	3.88%	2.06%	1.63%	2.50%
	LowerLipDepressor	11.63%	5.64%	3.01%	0.87%	0.80%	1.20%
	LipStretcher	7.84%	3.42%	2.32%	0.86%	1.06%	1.42%
	LipsToward	2.41%	2.13%	2.38%	2.15%	1.95%	2.21%
	LipPucker	0.67%	1.21%	2.38%	1.94%	1.10%	1.71%
	LipTightener	0.15%	0.42%	0.64%	0.68%	0.49%	0.60%
	MouthLeft	0.73%	0.48%	0.51%	0.31%	0.17%	0.42%
	LipCornerDepressor	0.43%	0.26%	0.58%	0.35%	0.39%	0.42%
	LipPressor	0.23%	0.15%	0.78%	0.27%	0.12%	0.40%
	LipFunneler	0.08%	0.16%	0.19%	0.33%	0.22%	0.20%
	MouthRight	0.05%	0.02%	0.03%	0.01%	0.02%	0.02%
Jaw	JawDrop	16.46%	13.57%	12.19%	10.24%	7.48%	9.76%
	JawThrust	1.68%	2.28%	3.19%	2.20%	0.68%	2.45%
	JawSidewaysRight	0.72%	0.44%	0.46%	0.26%	0.19%	0.31%
	JawSidewaysLeft	0.12%	0.13%	0.21%	0.14%	0.08%	0.12%

Joy, Surprise, and Disgust expressions exhibited greater amplitude, particularly in the Eyelid, Cheeks, and Mouth regions. These findings align with FACS-based studies [16, 54], which report larger facial movements for emotions involving smiling (Joy), mouth opening (Surprise), and squinting (Disgust). In contrast, Fear, Sadness, and Anger expressions were more subtle, characterized by micro-expressions [94].

The accuracy of these blendshape parameters is supported by their consistency with established literature on facial expressions. Furthermore, the use of high-fidelity facial tracking technology in the Meta Quest Pro headsets ensures precise capture of subtle facial movements [15], providing a reliable representation of the actors' expressions. This

combination of advanced hardware and alignment with theoretical frameworks reinforces the validity of our collected expression data.

4 STUDY 1: MODELING THE EFFECT OF FACIAL EXPRESSION SCALING

To systematically investigate the impact of facial expression scaling on users' perception of others' emotions and the uncanny valley effect in social VR applications, we conducted a controlled VR experiment. This study aimed to provide insights into how varying degrees of facial expression intensity influence emotional recognition and user comfort in virtual social interactions, addressing our first and second research questions:

- **RQ1:** How does facial expression scaling influence users' perception of others' emotions in social VR?
- **RQ2:** What is the relationship between facial expression scaling and the uncanny valley effect across different emotional expressions?

4.1 Study Design

We employed a 3×6 within-subjects design, where each participant experienced a series of pre-recorded emotional expressions in a virtual environment. These expressions varied across three scaling levels and six emotion categories. To ensure statistical rigor, we conducted an a priori power analysis [100], which indicated that a sample of 30 participants would achieve a statistical power of 0.8, assuming a medium effect size ($d = 0.5$) and an alpha error probability of 0.05.

The scaling levels were determined through a combination of literature review [44, 76, 104] and a pilot study. This approach helped establish values that were distinguishable in the experimental setting without being overly exaggerated or causing discomfort. As illustrated in Figure 1, We defined three scaling levels:

- **Natural:** Using the original facial blendshape strengths.
- **Damped:** Reducing the strength to 66.7% of the original.
- **Exaggerated:** Increasing the strength to 150% of the original.

Importantly, we adjusted all blendshapes except those related to gaze direction, as altering gaze-related blendshapes would have resulted in unnatural eye movements, potentially confounding our results.

Participants assumed the role of a listener in a simulated face-to-face conversation in social VR, experiencing 18 emotional expression clips. These clips were recorded from the same pair of actors and presented using gender-matched avatars to control for potential gender effects. To minimize order effects, we randomized the presentation of clips and ensured equal probability of assignment for each emotion across the three scaling conditions.

4.2 Measurements

To comprehensively assess the impact of scaled emotional expressions on listeners, we employed a multifaceted measurement approach, as summarized in Table 2. Participants completed a questionnaire after each recorded clip, encompassing both scaled measures and open-ended questions. The full questionnaire is available in the Supplementary Materials.

Our measurement framework was grounded in psychological theories of empathy and emotion perception. Following Powell and Roberts' [83] conceptualization of empathic responses, we evaluated two dimensions:

- **Cognitive Empathy:** Assessed through emotion recognition accuracy and perceived differences in emotion intensity [53]. This measures participants' ability to correctly identify and gauge the strength of expressed emotions.

- **Affective Empathy:** Measured via the degree of emotional contagion experienced by participants [17], indicating how effectively the avatar’s emotions were transmitted to the observer.

To evaluate the Uncanny Valley effect, we adopted scales widely used in previous studies [37, 38, 97]. These include vividness, appeal, realism, and eeriness of facial expressions, and overall comfort. Additionally, recognizing the importance of multimodal consistency in virtual interactions [66], we included a measure of perceived consistency between facial expressions and speech.

Table 2. Dependent variables collected in the questionnaire.

Emotion Perception	Emotion Recognition Accuracy: Recognition rate of six emotion categories (Joy, Sadness, Fear, Anger, Surprise, Disgust). Emotional Intensity Difference: Discrepancy between listener-assessed and speaker self-reported emotional intensity (7-point scale). Emotional Contagion: Listener’s self-reported empathy level (1: untouched, 7: highly empathetic).
Uncanny Valley Effect	Expression-Speech Consistency: Perceived alignment of facial expressions with speech (1: inconsistent, 7: consistent). Facial Vividness: Assessing the richness and clarity of facial expressions (1: bland and rigid, 7: rich and vivid). Facial Appeal: Measuring the attractiveness or likeability of facial expressions (1: unappealing, 7: appealing). Facial Realism: Gauging how true-to-life the expressions appeared (1: artificial, 7: realistic). Facial Eeriness: Evaluating any unsettling or disconcerting feelings evoked (1: normal, 7: eerie). Comfort: Assessing overall user comfort with the interaction (1: uncomfortable, 7: comfortable).
Open-ended Questions	Emotion Recognition Cues: Key indicators used for emotion identification. Expression-Speech Discrepancies: Noted misalignments between facial expressions and speech. Facial Expression Realism: Factors contributing to perceived realism or lack thereof. Facial Expression Appeal: Elements affecting attractiveness of expressions. Comfort Factors: Reasons behind comfort or discomfort during the experience.

This comprehensive set of measurements allows us to capture both the cognitive and affective aspects of emotion perception, as well as the nuanced effects of facial expression scaling on the uncanny valley phenomenon in social VR contexts.

4.3 Participants and Procedure

We recruited 30 Chinese participants (15 females, 15 males) from a university, aged 18 to 28 ($M = 23.2$, $SD = 2.3$). All participants were naive to the experiment’s objectives and had normal or corrected-to-normal vision. Participants’ familiarity with VR was assessed on a 5-point scale ($M = 2.97$, $SD = 1.13$). Six participants reported prior experience with social VR applications like VRChat.

Participants were instructed to imagine conversing with a friend in a social VR application, focusing on perceiving their friend’s emotions through observation and listening. The experiment consisted of 18 clips, with questionnaires

completed after each. Participants were divided into five groups, each experiencing expressions from different actor pairs. The order of scaling levels and emotion categories was counterbalanced to mitigate order effects. A 30-second break was provided between clips to prevent fatigue. The experiment lasted approximately 90 minutes, with participants receiving \$20 compensation. The study was approved by the university's ethical committee.

4.4 Analysis

We collected 540 questionnaire responses (18×30) with no outliers removed. Quantitative data were analyzed using one-way repeated measures ANOVAs, with Mauchly's test for sphericity and Greenhouse-Geisser corrections applied as necessary. Post-hoc comparisons used Bonferroni-corrected paired t-tests, with effect sizes reported as partial eta-squared (η_p^2).

For open-ended responses, we employed thematic content analysis [68] using incremental open coding [11]. The coding scheme (Table 3) was developed collaboratively by the research team. After initial coding by the primary researcher, a second team member independently reviewed the data, with disagreements resolved through team discussions.

Table 3. Description of the coding scheme identified in the thematic analysis of the qualitative data from the open-ended answers.

Theme	Category	Description
Emotion Recognition	Modalities	Communication channels (e.g., facial expressions, gaze, body movements, speech).
	Perceived Indicators	Observed emotional cues (e.g., melancholic voice, smile, bowed head).
Facial Features	Facial Region	Specific facial areas mentioned (e.g., eyebrows, cheeks, lips).
	Movements	Facial movements described (e.g., raise, open, frown).
	Scale	Extent of facial movements (e.g., insufficient, appropriate, excessive).
	Inconsistency	Misalignments between facial features and speech (e.g., lip sync, rhythm, emotion).
Description	Positive	Favorable descriptions (e.g., natural, appropriate, vivid).
	Negative	Unfavorable descriptions (e.g., uncanny, rigid, bland).

4.5 Quantitative Results

As summarized in Table 4, the main statistical results are as follows:

- Expression scaling did not significantly influence listeners' emotion perception.
- Expression scaling affected the uncanny valley effect: exaggerated expressions increased vividness but decreased facial realism, increased eeriness, and lowered comfort.
- Joy and sadness had higher recognition accuracy, while disgust more effectively triggered emotional contagion.
- The uncanny valley effect varied across emotions: joy and disgust showed higher expression-speech consistency and facial vividness; disgust exhibited higher facial realism and appeal compared to surprise; surprise and joy induced higher eeriness.
- A significant interaction effect was observed between expression scaling and emotional categories on uncanny valley indicators (facial appeal, realism, eeriness, and comfort).

Table 4. Summary of quantitative results. NS denotes nonsignificant results, while asterisk (*) indicates significant results (* at the < .05 level, ** at the < .01 level, and *** at the < .001 level). Values in parentheses represent pairwise comparison results: D stands for Damped, N stands for Natural, and E stands for Exaggerated. Jo, Sa, Fe, An, Su, and Di refer to Joy, Sadness, Fear, Anger, Surprise, and Disgust, respectively.

Measurements		Main Effect of Expression Scaling	Main Effect of Emotion Category	Interaction Effect
Emotion Perception (RQ1)	Emotion Recognition Accuracy	NS	*** (Su, Di, Fe, An < Jo; Di, Fe, An < Sa)	NS
	Emotional Intensity Difference	NS	**	NS
	Emotional Contagion	NS	** (Fe, Su < Di)	NS
Uncanny Valley Effect (RQ2)	Expression-Speech Consistency	NS	*** (Su, Fe, Sa, An < Jo; Su, Fe < Di)	NS
	Facial Vividness	** (D < E)	*** (Fe, Sa, An < Jo; Fe, Sa, An < Di)	NS
	Facial Appeal	NS	** (Su < Di)	**
	Facial Realism	* (E < N)	* (Su < Di)	**
	Facial Eeriness	*** (E < D, N)	*** (Di, Sa, An < Su, Sa < Jo)	***
	Comfort	** (E < D, N)	NS	**

These findings provide insights into the complex relationship between facial expression scaling, emotion types, and user perceptions in social VR contexts. The following are the specific statistical analysis results.

4.5.1 Emotion Perception. Expression Scaling has no significant effect on users' emotion perception (Figure 4). However, despite the lack of statistical significance, mean values suggest that increasing the expression scaling slightly enhances emotion recognition accuracy, from Damped ($M = 0.83$, $SD = 0.15$) to Natural ($M = 0.84$, $SD = 0.11$) to Exaggerated ($M = 0.86$, $SD = 0.15$). Similarly, for Emotional Intensity Difference, higher expression scaling leads listeners to judge the speaker's emotional intensity more strongly, from Damped ($M = 0.16$, $SD = 0.17$) to Natural ($M = 0.33$, $SD = 0.19$) to Exaggerated ($M = 0.51$, $SD = 0.14$). As for Emotional Contagion, there is virtually no difference among Damped ($M = 4.99$, $SD = 0.16$), Natural ($M = 4.97$, $SD = 0.16$), and Exaggerated ($M = 4.96$, $SD = 0.17$).

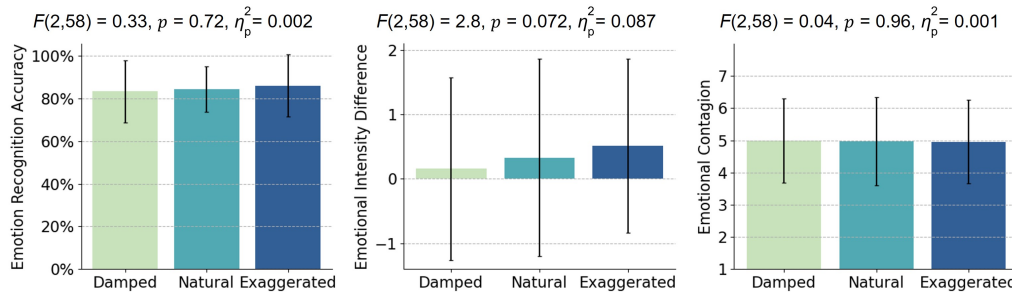


Fig. 4. The main effect of Expression Scaling on Emotion Recognition Accuracy (left), Emotional Intensity Difference (middle), and Emotional Contagion (right). The error bars denote standard deviation.

Moreover, Emotion Category has a significant impact on users' emotion perception (Figure 5). Regarding Emotion Recognition Accuracy, the recognition rate for Joy ($M = 1.0$, $SD = 0.0$) is significantly higher than for Surprise

($M = 0.83, SD = 0.27, p < .05$), Disgust ($M = 0.81, SD = 0.24, p < .01$), Fear ($M = 0.73, SD = 0.22, p < .0001$), and Anger ($M = 0.72, SD = 0.26, p < .0001$), whereas the recognition rate for Sadness ($M = 0.98, SD = 0.08$) is significantly higher than for Disgust ($p < .05$), Fear ($p < .0001$), and Anger ($p < .0001$). This pattern is consistent with previous studies [36, 75, 81, 105], which have found that happiness and sadness are more easily recognized in both real and cartoon faces. The higher recognition rates for these emotions may be due to their distinct and culturally universal expression patterns.

For Emotional Intensity Difference, there are no significant pairwise differences observed. In terms of Emotional Contagion, Disgust ($M = 5.43, SD = 0.16$) is significantly higher than Fear ($M = 4.61, SD = 0.20, p < .001$) and Surprise ($M = 4.70, SD = 0.16, p < .001$). This pattern may be attributed to its pronounced facial expressions (e.g., squinting and frowning) and associated social signals (e.g., “Ugh” sounds), which collectively create a more visceral and transmissible emotional experience.

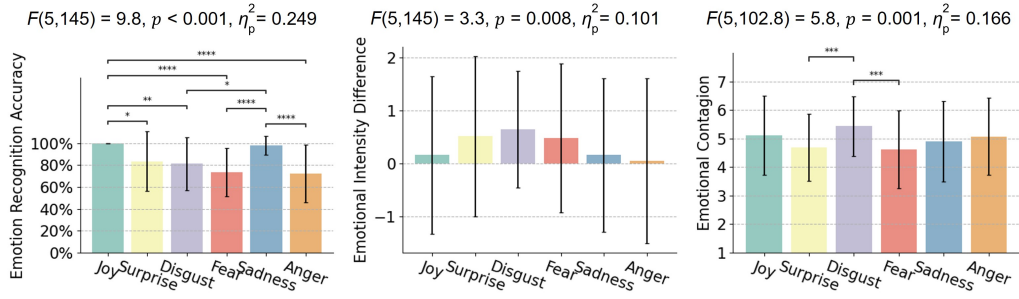


Fig. 5. The main effect of Emotion Category on Emotion Recognition Accuracy (left), Emotional Intensity Difference (middle), and Emotional Contagion (right). The error bars denote standard deviation, and *, **, ***, and **** denote significant differences at $p < 0.05$, $p < 0.01$, $p < 0.001$, and $p < 0.0001$ levels, respectively.

The analysis of emotion recognition confusion (Figure 6) reveals specific patterns of misidentification across emotions, regardless of expression scaling levels. Notably, Disgust was often mistaken for Anger, Fear for Surprise and Disgust, and Anger for Disgust. These findings align with previous research on emotion recognition challenges [27, 86, 87], suggesting that certain emotion pairs are more easily confused due to similarities in their facial expressions. The lack of significant interaction effects between Expression Scaling and Emotion Category for recognition accuracy, intensity difference, and emotional contagion indicates that scaling impacts these aspects consistently across emotions.

4.5.2 Uncanny Valley Effect. Expression Scaling significantly affects several uncanny valley indicators (Figure 7). Specifically, Facial Vividness in the Exaggerated condition ($M = 4.94, SD = 0.18$) is significantly greater than in the Damped condition ($M = 4.36, SD = 0.16, p < .01$). However, Facial Realism in the Exaggerated condition ($M = 4.33, SD = 0.15$) is significantly lower than in the Natural condition ($M = 4.71, SD = 0.15, p < .05$). Additionally, Facial Eeriness in the Exaggerated condition ($M = 3.74, SD = 0.13$) is significantly higher than in both the Damped ($M = 3.26, SD = 0.14, p < .01$) and Natural conditions ($M = 3.19, SD = 0.11, p < .0001$), while Comfort in the Exaggerated condition ($M = 4.39, SD = 0.15$) is significantly lower than in both the Damped ($M = 4.76, SD = 0.17, p < .05$) and Natural conditions ($M = 4.81, SD = 0.14, p < .001$).

Moreover, Emotion Category significantly affects several uncanny valley indicators (Figure 8). For Expression-Speech Consistency, Joy ($M = 5.58, SD = 0.15$) is significantly higher than Surprise ($M = 4.66, SD = 0.17, p < .01$), Fear

Scripted Emotion	Joy	90	0	0	0	0	0
	Surprise	4	75	4	0	0	2
	Disgust	0	0	73	3	0	14
	Fear	1	7	9	66	1	4
	Sadness	0	0	0	0	88	0
	Anger	0	1	20	0	1	65
		Perceived Emotion					
		Joy	Surprise	Disgust	Fear	Sadness	Anger

Fig. 6. Confusion matrix illustrating emotion recognition rates for six basic emotions, with the three expression scaling levels considered together.

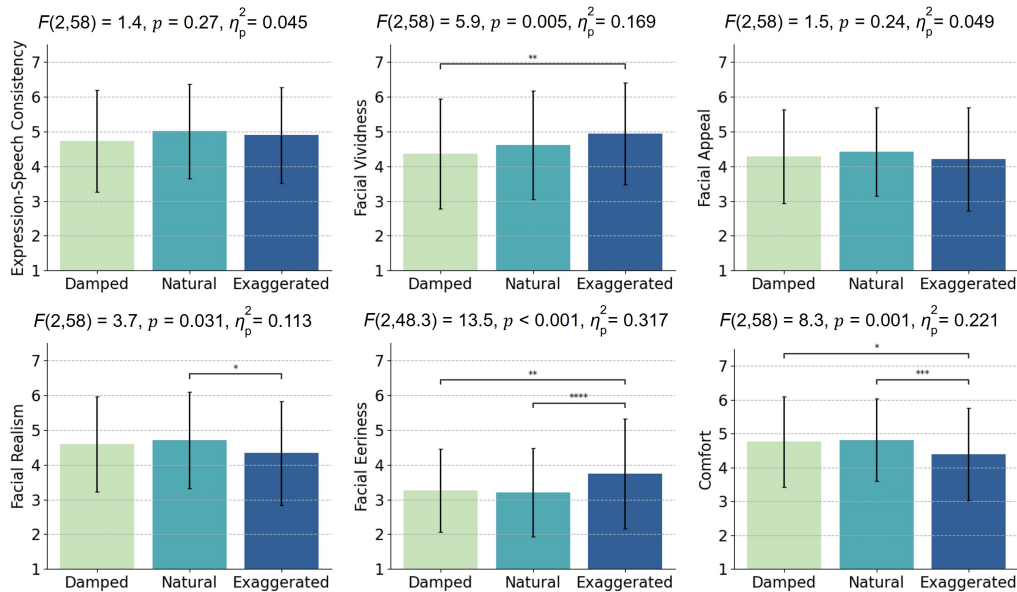


Fig. 7. The main effect of Expression Scaling on Expression-Speech Consistency (top left), Facial Vividness (top middle), Facial Appeal (top right), Facial Realism (bottom left), Facial Eeriness (bottom middle), and Comfort (bottom right). The error bars denote standard deviation, and *, **, ***, and **** denote significant differences at $p < 0.05$, $p < 0.01$, $p < 0.001$, and $p < 0.0001$ levels, respectively.

($M = 4.44, SD = 0.17, p < .0001$), Sadness ($M = 4.71, SD = 0.17, p < .01$), and Anger ($M = 4.71, SD = 0.18, p < .01$); Disgust ($M = 5.14, SD = 0.18$) is significantly higher than Surprise ($p < .05$) and Fear ($p < .05$). Regarding Facial Vividness, Joy ($M = 5.26, SD = 0.21$) is significantly greater than Fear ($M = 4.41, SD = 0.19, p < .001$), Sadness

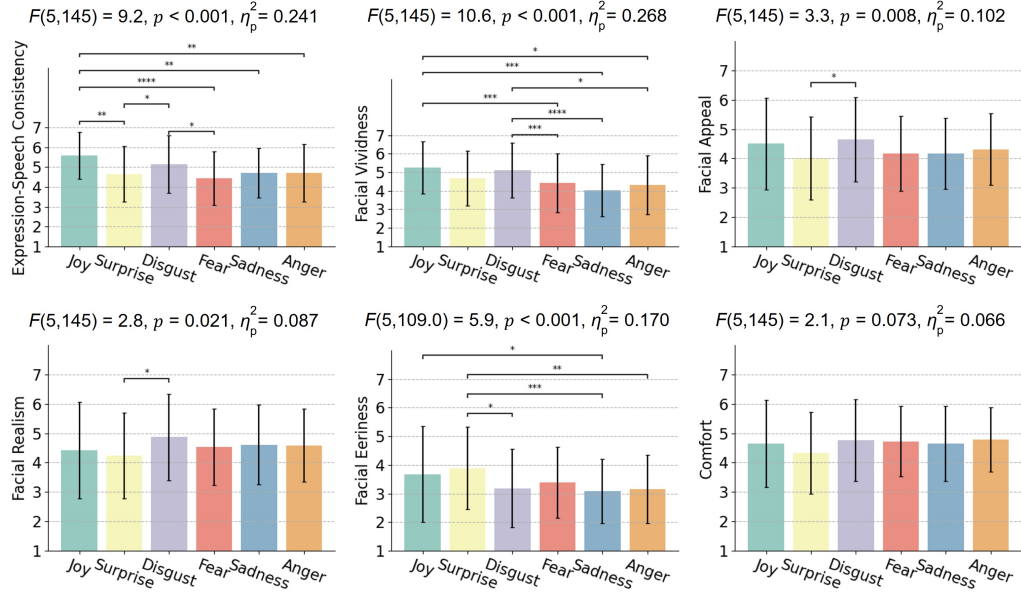


Fig. 8. The main effect of Emotion Category on Expression-Speech Consistency (top left), Facial Vividness (top middle), Facial Appeal (top right), Facial Realism (bottom left), Facial Eeriness (bottom middle), and Comfort (bottom right). The error bars denote standard deviation, and *, **, ***, and **** denote significant differences at $p < 0.05$, $p < 0.01$, $p < 0.001$, and $p < 0.0001$ levels, respectively.

($M = 4.03, SD = 0.18, p < .001$), and Anger ($M = 4.32, SD = 0.19, p < .05$), while Disgust ($M = 5.11, SD = 0.19$) is significantly greater than Fear ($p < .001$), Sadness ($p < .0001$), and Anger ($p < .05$). For Facial Appeal, Disgust ($M = 4.64, SD = 0.19$) is significantly higher than Surprise ($M = 4.17, SD = 0.15, p < .05$). Similarly, for Facial Realism, Disgust ($M = 4.87, SD = 0.19$) is significantly higher than Surprise ($M = 4.23, SD = 0.17, p < .05$). Concerning Facial Eeriness, Joy ($M = 3.68, SD = 0.20$) is significantly higher than Sadness ($M = 3.09, SD = 0.13, p < .05$); Surprise ($M = 3.89, SD = 0.14$) is significantly higher than Disgust ($M = 3.18, SD = 0.17, p < .05$), Sadness ($p < .001$), and Anger ($M = 3.16, SD = 0.16, p < .01$).

Interestingly, significant interaction effects were observed between Expression Scaling and Emotion Category for several uncanny valley indicators (Figure 9). These interactions suggest that the impact of expression scaling on user perception varies depending on the specific emotion being conveyed. For instance, increased scaling negatively affected the perceived appeal and realism of Surprise, Joy, and Disgust expressions, while positively impacting Anger, Fear, and Sadness. This divergence in effects across emotions highlights the complexity of facial expression perception in virtual environments and underscores the need for emotion-specific approaches in avatar design.

4.6 Qualitative Results

In Table 5, we summarize the primary non-facial and facial indicators that participants reported using for emotion recognition in their open-ended answers, as well as their main evaluations for damped and exaggerated facial expressions.

Participants perceive emotions through various modalities, including speech content, tone, facial expressions, and body posture. Statistical analysis revealed that facial expressions account for only 33.2% of the mentioned cues, while 66.8% are non-facial indicators. Among these, speech content and tone were the most frequently cited factors (37.1%).

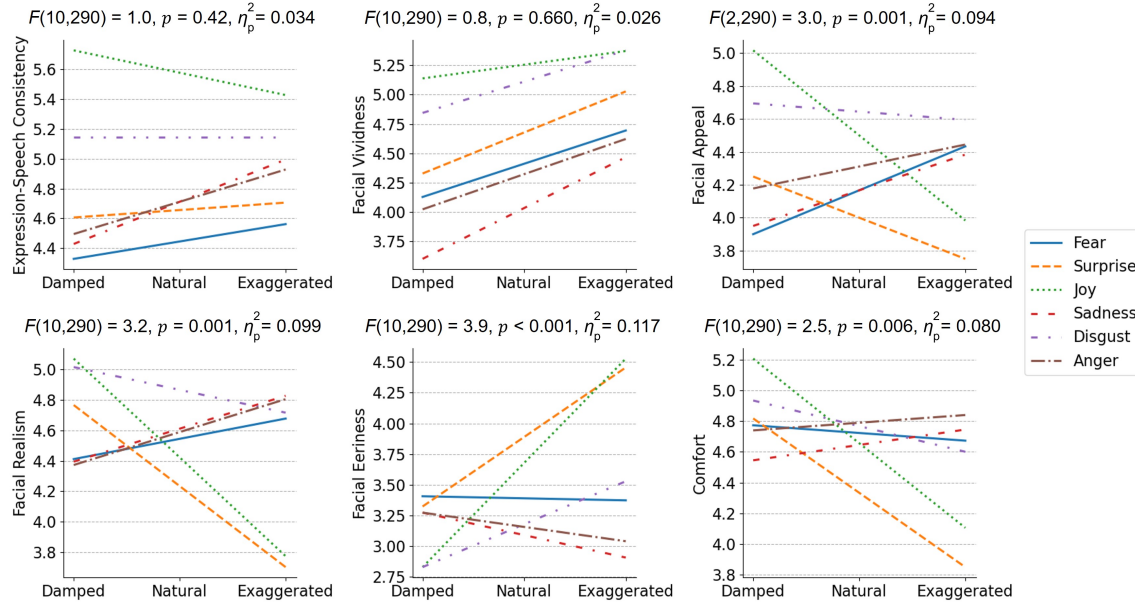


Fig. 9. The interaction effect of Emotion Category $\times$ Expression Scaling on Expression-Speech Consistency (top left), Facial Vividness (top middle), Facial Appeal (top right), Facial Realism (bottom left), Facial Eeriness (bottom middle), and Comfort (bottom right).

Facial expressions are often considered secondary in emotion judgment. As indicated by P17, “I know the avatar in front of me is just a synthesized virtual figure, and their expressions may not be accurate, whereas the voice I hear is real. Therefore, I rely more on the speech to determine their emotions, with facial expressions serving as an auxiliary cue.”

The effects of Damping and Exaggerating facial expressions varied significantly across different emotion types. For Joy and Surprise, participants noted that Exaggerating could make the expressions overly exaggerated and unnatural, particularly the movements of the mouth. When evaluating an exaggerated Joy clip, P5 remarked, “The speaker smiles so broadly that they show all their upper gums and even the inside of their mouth, which is very unsettling.” Similarly, exaggerated expressions for Surprise often resulted in overly wide-open mouths, excessive cheek raising, and facial proportion imbalances. As P7 commented, “The speaker’s cheek movements are so exaggerated that they squint their eyes into slits, which I found somewhat creepy.” Additionally, exaggerated mouth movements can cause issues with lip synchronization. As P14 pointed out, “The speaker’s mouth movements and their speech content seem mismatched. They are keeping their mouth wide open in a big smile while speaking, which feels very eerie.”

In contrast, the Natural and Damped conditions for Joy and Surprise expressions were described as more “natural” and “believable.” P17 mentioned that the Damped Joy clip made her feel comfortable, “The speaker’s smile causes their cheeks to push the eyes into a curved shape, showing that she is truly happy. Her smile feels very natural, like that of a real person.”

For Fear, Sadness, and Anger, participants generally noted that exaggerated expressions were more “recognizable,” “vivid,” and conveyed clearer emotions. After observing an exaggerated Fear clip, P27 pointed out, “The speaker’s brows are furrowed tightly, and her eyes convey emotions of fear and anxiety. The micro-expressions are realistic and rich.” Similarly, P28, after viewing an exaggerated Anger clip, remarked, “I can clearly see the muscle movements in the speaker’s

Table 5. Thematic analysis summary of open-ended answers. For different emotion categories, the second column lists the non-facial and facial indicators referenced by participants for emotion recognition, with the number of mentions in parentheses (only indicators mentioned more than three times are listed in the table). The third and fourth columns summarize the main evaluations of facial expressions under Damped and Exaggerated conditions, respectively. Positive evaluations are highlighted in **blue**, and negative evaluations are highlighted in **red**.

Emotions	Emotional Indicators	Eval. of Damped FE	Eval. of Exaggerated FE
Joy	Non-facial: <i>speech and tone</i> (82), <i>laughter</i> (40), <i>faster pace</i> (12), <i>raised arms</i> (7) Facial: <i>smile</i> (36), <i>mouth corners lifting</i> (23), <i>raising of the cheeks</i> (21), <i>curved eyes</i> (8).	Natural expression (P5, P12, P14, P17, P18, P21), coordinated movements across different facial regions (P3, P15, P18, P21), look nice (P12, P13, P20, P25).	Overexaggerated smile (P2, P5, P7, P8, P14, P16, P28), overly wide mouth opening (P2, P7, P8, P13, P21, P24), disproportionately high cheek raising (P4, P14, P20, P24).
Surprise	Non-facial: <i>speech and tone</i> (78), <i>sharp intake of breath</i> (19), <i>"Oh!" exclamations</i> (15) Facial: <i>widely open mouth</i> (27), <i>raised eyebrows</i> (19), <i>wide-open eyes</i> (14).	Natural expression (P3, P18, P23, P28, P29), believable appearance (P7, P23, P26, P27), appropriate amplitude of motion (P18, P26).	Overly open jaw (P1, P8, P9, P10, P12, P14, P25, P30), excessively dramatic expression changes (P4, P10, P22, P26), disproportionately high cheek raising (P5, P7, P15, P18).
Disgust	Non-facial: <i>speech and tone</i> (76), <i>"Ugh" sounds</i> (21), <i>hands pushing outward</i> (6) Facial: <i>wrinkled nose</i> (30), <i>furrowed brows</i> (20), <i>downturned mouth corners</i> (12), <i>pursed lips</i> (7), <i>squinting eyes</i> (5).	Lacking facial movement (P2, P4, P6, P10, P12), a lower emotional intensity than speech (P8, P21, P24, P28), realistic brows frowning (P17, P23, P28), natural and vivid eye squint (P3, P6).	Overly open mouse (P6, P8, P16, P22, P27, P29), unnatural eye squint (P2, P3, P8, P22, P25), noticeable nose wrinkling (P1, P5, P10, P19, P27), dynamic and expressive cheek raising (P4, P26, P29).
Fear	Non-facial: <i>speech and tone</i> (66), <i>trembling</i> (20), <i>raised shoulders</i> (12) Facial: <i>raised brows</i> (17), <i>widened eyes</i> (9), <i>trembling lips</i> (4), <i>darting gaze</i> (3).	Stiff facial expression (P10, P20, P22, P23, P27, P28, P30), limited movement with only noticeable mouth opening and closing (P1, P4, P7, P9, P11, P13).	Rich micro-expressions (P1, P2, P18, P25, P27), noticeable furrowed brows (P17, P25, P26, P27), trembling facial muscles (P20, P21, P25), rich micro-expressions (P11, P19, P29).
Sadness	Non-Facial: <i>speech and tone</i> (78), <i>head down</i> (40), <i>slower pace</i> (36), <i>sighing</i> (18) Facial: <i>downward gaze</i> (25), <i>downturned mouth corners</i> (18), <i>drooping eyebrows</i> (9).	Blank face (P4, P8, P13, P25, P26), rigid facial expressions (P1, P11, P12, P16, P21), a tiny smile that contradicted the emotion (P1, P5, P7, P27, P30).	Vivid and authentic expression (P1, P3, P22, P28, P30), Noticeable eyebrows drooping (P4, P17, P19, P22, P29), natural lid squinting (P18, P22, P29, P30).
Anger	Non-facial: <i>speech and tone</i> (74), <i>rapid pace</i> (36), <i>aggressive posture</i> (13), <i>clenched fists</i> (7) Facial: <i>glare</i> (29), <i>furrowed brows</i> (28), <i>bared teeth</i> (8), <i>raising of the cheeks</i> (5).	Flat, stiff expression with limited facial movement (P2, P4, P20, P21, P22, P25, P26, P30), a lower emotional intensity than speech (P7, P8, P21, P28).	Obvious nose wrinkling (P4, P8, P9, P10, P20, P28), naturally furrowed brows (P14, P15, P23, P24, P27), expressive stare (P12, P18, P22), vivid cheek raising (P22, P25, P28).

face, their face scrunched up, which strongly conveys their anger." Participants also noted that eyebrow movements are

crucial for conveying negative emotions and suggested further increasing the magnitude of these movements (P14, P17, P22, P25, P28).

Conversely, participants described the facial expressions for Fear, Sadness, and Anger under the Damped and Natural conditions as “stiff,” “bland,” “lacking emotion,” and “missing micro-expressions.” Critiquing a Damped Fear clip, P13 stated, “*The speaker’s mouth does not move much while talking, and there are no other expressions on their face. It feels like only their mouth is mechanically opening and closing, which does not align with their tone.*” P18 criticized a Natural Sadness clip, saying, “*The speaker is talking about something very sad, but she seems almost expressionless. Even her lips are slightly upward, making her look somewhat happy, which confuses me and makes me distrust her expression.*” P4 commented on a Damped Anger clip, “*A normal person, when very angry, would frown and wrinkle their nose. However, I noticed that the speaker’s cheeks were stiff and immobile; the upper facial muscles showed no reaction while speaking, which felt very unrealistic to me.*”

For Disgust, participants provided mixed evaluations for both the Damped and Exaggerated conditions. In the Damped condition, while the expressions lacked activity in other facial regions, participants appreciated the realistic frowning and natural squinting. Conversely, in the Exaggerated condition, the more pronounced nose wrinkling and cheek raising were noted to enhance the richness of the expression. As P1 commented, “*I can see the speaker squinting their eyes, and when they look up and down, it feels as if they are rolling their eyes, which is very vivid.*” However, overly exaggerated mouth opening and squinting were also criticized.

4.7 Discussion

4.7.1 The Effect of Facial Expression Scaling on Emotion Perception (RQ1). Our study addresses a significant gap in the literature by examining the effects of facial expression scaling on emotion perception in social VR contexts. Unlike previous research that primarily focused on static expressions [12] or 2D screen-based interactions [96], our study simulates realistic face-to-face interactions in immersive VR environments, incorporating dynamic facial expressions, speech, gaze, and gestures [67]. This approach provides a more comprehensive understanding of emotional communication in social VR.

Surprisingly, our analysis (Figure 4) reveals that neither damping nor exaggerating facial expressions significantly impacted emotion recognition accuracy, intensity perception, or emotional contagion. This finding challenges the assumption that enhanced facial expressions necessarily lead to improved emotional communication in virtual environments. The primary reason for this unexpected result, as indicated by 25 out of 30 participants, is their preference for vocal cues and speech content as the predominant factors in emotion judgment. Additionally, participants mentioned non-facial cues approximately two-thirds of the time, while facial expressions accounted for only one-third.

Our findings align with previous research highlighting the primacy of vocal cues in multimodal emotional information processing during face-to-face interactions [2, 42]. Furthermore, they support studies emphasizing the importance of sound [43] and body movements [77] as crucial channels for emotion recognition in social VR. This suggests that despite the vital role of avatar facial expressions in enhancing immersion, embodiment, and social presence in social VR [21, 28, 52], their impact on emotional communication may be more limited than previously thought.

An additional factor contributing to this result is the perceived authenticity of voices compared to virtual facial expressions. Participants reported greater trust in genuine vocal cues over synthesized visual expressions, leading to reduced reliance on avatar facial expressions. These results highlight the complex nature of emotional communication in social VR and underscore the need for a holistic approach to avatar design, suggesting that improvements in facial expression fidelity should be balanced with enhancements in other communication channels, particularly voice and

speech content. Future research should focus on optimizing the interplay between different modalities to create more effective and engaging social VR experiences.

4.7.2 The Effect of Facial Expression Scaling on Uncanny Valley Effect (RQ2). Our study addresses significant gaps in the existing literature on facial expression scaling in virtual environments, particularly within the context of social VR. While previous research [31, 44, 66, 76] has explored facial expression manipulation, it has not adequately considered the effects across different emotion types in immersive social contexts.

Our findings reveal a complex relationship between facial expression scaling and the uncanny valley effect. Contrary to some existing studies [44, 76], we found that exaggeration does not uniformly enhance user experience. Instead, we observed significant variations in the impact of expression exaggeration across different emotions (Figure 9). For Surprise, Joy, and Disgust, exaggerated expressions intensified the uncanny valley effect, aligning with the theory that overly human-like or exaggerated features can elicit discomfort [66]. Conversely, for Fear, Sadness, and Anger, facial exaggeration appeared to mitigate the uncanny valley effect, enhancing realism and appeal. This emotion-dependent effect is a novel finding that challenges the notion that more realistic or consistently scaled expressions are always preferable.

Our analysis of blendshape activations (Table 1) provides insights into these differential effects. Joy, Surprise, and Disgust, characterized by larger mouth and cheek movements [5, 29], become visually discomforting when further exaggerated. In contrast, Fear, Sadness, and Anger, typically expressed through subtler eyebrow, eyelid, and nose movements, benefit from exaggeration. This finding is consistent with previous research indicating that a lack of facial expressions in virtual characters can trigger the uncanny valley effect [95].

We posit that the subtle nature of negative emotions' expressions is less discernible on cartoon avatars compared to real human faces [64, 73]. Additionally, the physical constraints of head-mounted displays may limit upper facial muscle movements [9, 28], further reducing the visibility of these subtle expressions. Exaggeration of these initially subtle movements makes them more recognizable, leading to increased perceived naturalness and realism.

Our findings collectively indicate that a nuanced, facial-region-specific approach to expression scaling in virtual avatars is crucial for optimizing emotional communication while mitigating uncanny valley effects. This insight challenges the prevalent one-size-fits-all approach in avatar facial enhancement strategies. Future design strategies should consider the differential impacts of scaling across various facial regions, such as the upper face (eyebrows, eyelids) and lower face (mouth, cheeks). By tailoring expression scaling to specific facial areas, we can potentially enhance the clarity and effectiveness of emotional communication in virtual environments while maintaining user comfort and perceived naturalness. We also propose that adaptive scaling algorithms could be developed to adjust the degree of exaggeration based on the specific emotion being conveyed and the context of the interaction.

5 STUDY 2: VALIDATION OF SELECTIVE EXAGGERATION STRATEGY

Building upon the insights gained from Study 1, we identified a nuanced relationship between facial expression scaling and the uncanny valley effect across different emotions. Our findings revealed that the impact of exaggeration varies significantly depending on the emotion and the specific facial regions involved. These observations suggested that a more nuanced, facial-region-specific approach to facial expression enhancement might be beneficial for optimizing emotional communication in virtual avatars while mitigating potential uncanny valley effects.

Building on these insights, we developed a selective facial expression exaggeration strategy. This approach aims to amplify the less pronounced facial features while maintaining the natural intensity of already expressive elements. The

primary motivation for this second study was to validate the effectiveness of this refined strategy and to address our third research question:

- **RQ3:** How can we leverage facial expression scaling to enhance emotional communication in social VR while mitigating potential uncanny valley effects?

5.1 Selective Facial Exaggeration Strategy

Our findings from Study 1 revealed a nuanced relationship between facial expression exaggeration and the uncanny valley effect, varying across different facial regions and emotions. Exaggerating the lower facial region—particularly the mouth and jaw—induced the uncanny valley effect for Joy and Surprise, likely because these areas are already naturally expressive for these emotions. Conversely, amplifying the upper facial region, including the eyebrows, eyelids, and nose, enhanced the expressiveness and perceived realism of Fear, Sadness, and Anger. This effect is probably due to these features being typically under-expressive in virtual environments, especially when constrained by head-mounted displays.

Eyebrows, in particular, play a crucial role in conveying emotions [95, 103]. Interestingly, none of our participants reported the eyebrow movements as overly exaggerated; some even suggested further amplification to enhance expressiveness. This indicates a potential "sweet spot" for eyebrow exaggeration that enhances emotional conveyance without triggering the uncanny valley effect.

The impact of cheek exaggeration varied by emotion: while it resulted in unnatural expressions for Joy and Surprise, it improved expressiveness for Disgust and Anger. This variation suggests that cheek movements contribute differently to different emotional expressions and should be treated accordingly in avatar design.

Based on these insights, we propose a selective facial expression exaggeration strategy, as outlined in Table 6. This approach aims to optimize emotional expressiveness while mitigating the risk of the uncanny valley effect:

Table 6. Specific settings for selective facial expression exaggeration strategy.

Facial regions	Exaggeration level	Scale
Eyebrows	Highly Exaggerated	200%
Eyelid, Nose	Moderately Exaggerated	150%
Cheeks	Weakly Exaggerated	125%
Gaze, Mouth, Jaw	Original	100%

5.2 Study Design

To validate the effectiveness and applicability of our proposed selective facial expression exaggeration strategy across different emotional categories, we conducted a comparative study. This experiment utilized the same virtual environment and emotional expression materials as Study 1, ensuring consistency and allowing for direct comparison of results.

Participants assumed the role of listeners in a simulated face-to-face VR interaction. They were presented with gender-matched avatar speakers displaying facial expressions, body postures, and eye movements recorded from actors, accompanied by the actors' real voices. This setup aimed to create a realistic and immersive social VR experience.

For each emotional scenario, participants viewed two versions of the same clip in randomized order: one with selectively exaggerated facial expressions according to our strategy, and one with unmodified original expressions. This pairwise comparison approach allowed for direct evaluation of the impact of our selective exaggeration strategy.

After viewing each pair of clips, participants completed a questionnaire. Based on the findings from Study 1, which showed that expression scaling did not significantly affect emotion recognition accuracy, we focused our assessment on other aspects of emotion perception and the uncanny valley effect. Participants rated eight dimensions: emotional intensity, emotional contagion, expression-speech consistency, facial vividness, facial appeal, facial realism, facial eeriness, and overall comfort. For each dimension, they indicated their preference using a 5-point Likert scale, responding to the statement "The second clip is better than the first one" (ranging from "Strongly disagree" to "Strongly agree"). To gain deeper insights into participants' preferences and perceptions, we also included open-ended questions. These allowed participants to elaborate on the reasons behind their choices, providing valuable qualitative data to complement our quantitative measures.

5.3 Participants and Procedure

We recruited 20 participants (12 females, 8 males, age range: 18-29 years, $M = 24.0$, $SD = 2.9$) from a Chinese university campus. This sample size was chosen to provide sufficient statistical power (0.8) for our within-subjects design while remaining feasible for the in-depth, VR-based nature of the study. All participants had normal or corrected-to-normal vision and were naive to the study's objectives, ensuring unbiased responses. None had participated in our previous experiments, avoiding potential carry-over effects. To gauge participants' prior experience, we asked them to rate their familiarity with VR on a 5-point scale ($M = 2.83$, $SD = 1.25$). Additionally, we inquired about their experience with social VR applications; three out of 20 participants reported using platforms like VRChat. This information helps contextualize our findings within the broader landscape of VR user experience.

Before the experiment, participants received the following instructions: "Imagine you are conversing with a friend through a social VR application, and your friend is sharing their personal experiences with you. Your task is to listen to your friend's expressions and observe their facial expressions. You will observe 9 pairs of virtual human expression clips in VR. Each pair includes two observations where the content is the same but the facial expressions have slight differences." This framing was designed to create a realistic social VR scenario while focusing participants' attention on the facial expressions, which were the key variable of interest.

Each participant viewed all 9 emotion clips from one of 10 actors, with each actor's clips randomly assigned to two participants. This design ensured a balanced exposure to different actors while maintaining a manageable experiment duration. The order of clip presentation was randomized to mitigate potential order effects. Participants completed questionnaires after each pair of clips, with a 30-second break between videos to prevent fatigue. The entire experiment lasted approximately 90 minutes. Participants received \$20 compensation. The study protocol was approved by the local university's ethics committee.

5.4 Analysis

We collected 180 paired comparison results (20 participants $\times$ 9 emotion clips), retaining all data points to maintain the integrity of our dataset. To ensure consistent analysis, we standardized all ratings to address the question: "Are the selectively exaggerated facial expressions better than the unmodified original ones?" This standardization involved subtracting the rating for the unmodified original version from that of the selectively exaggerated version for each pair.

Our quantitative analysis followed a rigorous statistical approach: (1) We first conducted Shapiro-Wilk tests to assess the normality of distribution for each measure. (2) For normally distributed data, we used One-Sample T-tests to compare the mean rating against the neutral point of 3 ("neither agree nor disagree") on our 5-point Likert scale. (3) For non-normally distributed data, we applied the non-parametric Mann-Whitney U test as an alternative, still comparing against

the neutral point of 3. This method allowed us to determine whether participants showed a statistically significant preference for either the selectively exaggerated or unmodified original expressions across various measures of emotion perception and uncanny valley effects.

For the qualitative data from open-ended responses, we employed the same thematic content analysis method used in Study 1. This consistency in qualitative analysis techniques facilitates direct comparisons between insights gained from both studies.

5.5 Results

5.5.1 Quantitative Results. The quantitative analysis of our selective facial expression exaggeration strategy revealed varying effects across different emotions and dimensions of user experience. Figure 10 presents bar charts depicting the distribution of participants' ratings across different emotional categories and various uncanny valley dimensions. Our strategy proved most effective for Disgust, Fear, and Anger expressions. For Disgust, participants significantly preferred the selectively exaggerated expressions across all measured dimensions (all $p < .05$), with notable improvements in realism ($M = 3.72$, $SD = 0.96$, $p < .001$) and speech consistency ($M = 3.62$, $SD = 0.86$, $p < .001$). Fear expressions showed significant improvements in all dimensions except Facial Eeriness, with Facial Vividness ($M = 3.73$, $SD = 1.08$, $p < .001$) and Emotional Contagion ($M = 3.63$, $SD = 1.16$, $p < .01$) showing the most pronounced enhancements. Anger expressions demonstrated the strongest preference for selective exaggeration across all dimensions (all $p < .01$), particularly in Emotional Contagion ($M = 3.87$, $SD = 0.82$, $p < .0001$), Facial Realism ($M = 3.93$, $SD = 0.94$, $p < .0001$), and overall Comfort ($M = 4.00$, $SD = 0.95$, $p < .0001$).

Interestingly, for Sadness, only Facial Realism showed significant improvement ($M = 3.47$, $SD = 1.22$, $p < .05$), while Joy and Surprise expressions exhibited no statistically significant differences between the selectively exaggerated and original versions. These results suggest that our strategy is particularly effective in enhancing the perception and impact of negative emotions, especially those typically associated with subtle facial expressions, while maintaining or slightly improving the experience for positive emotions.

5.5.2 Qualitative Results. For negative emotions such as Fear, Disgust, and Anger, participants consistently expressed a preference for the selectively exaggerated expressions. They reported that these enhanced expressions aligned better with the conveyed emotional content, resulting in richer, more natural, and more intense facial displays. Participants frequently mentioned that the more pronounced frowning, glaring, squinting, and cheek raising contributed to a stronger and more realistic emotional conveyance that felt more consistent with the accompanying speech (P3, P5, P10, P11, P12, P13, P15, P19). The exaggeration of eyebrow movements was particularly effective for Disgust and Anger, as exemplified by P6's comment: "The frown in the second (selectively exaggerated) clip is more real, like staring at someone, the emotion is stronger and more real."

In contrast, for positive emotions like Joy and Surprise, as well as for Sadness, participants often reported difficulty in detecting differences between the original and selectively exaggerated clips. This observation is consistent with our quantitative findings. However, some participants did note potential issues with the exaggerated eyebrow and eyelid movements for Joy and Surprise. For instance, P14 observed that more pronounced eyebrow movements in joyful expressions created a sense of frowning, which felt incongruent with the emotion. Similarly, P8 expressed discomfort with the exaggerated squinting in Surprise expressions, stating, "In the second (selectively exaggerated) clip, the speaker's eyes looked like slits, almost mocking, which felt eerie." These observations highlight the delicate

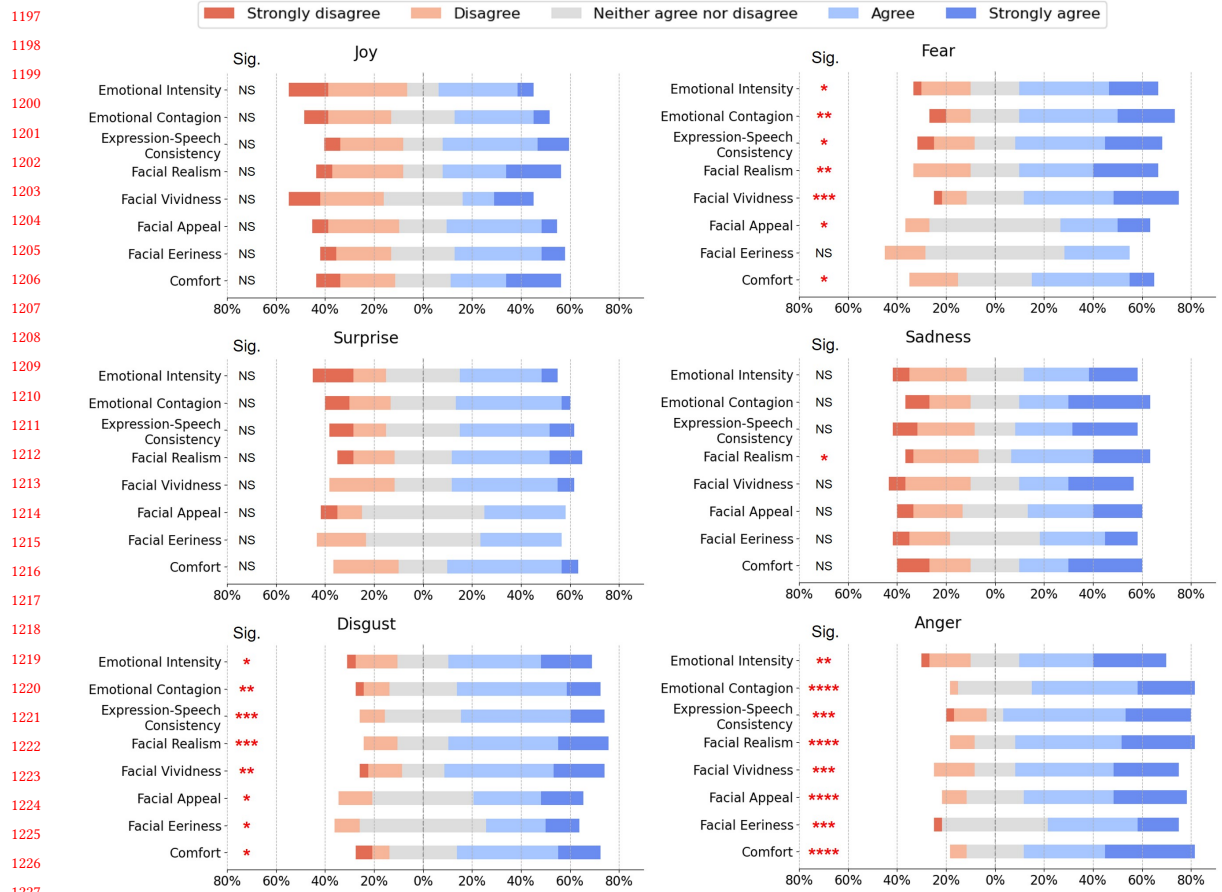


Fig. 10. Participants' ratings for the question: "Are the perception of selectively exaggerated facial expressions better (or stronger) than the unmodified original ones?" Results are categorized by emotional categories and various dimensions of emotion perception and uncanny valley effect. Statistical tests were conducted to compare the mean rating against the neutral point. Significance levels are indicated as follows: * $p < .05$, ** $p < .01$, *** $p < .001$, **** $p < .0001$. NS indicates no statistically significant difference.

balance required in enhancing positive emotional expressions and suggest areas for further refinement in our selective exaggeration strategy.

5.6 Discussion: Leveraging Selective Facial Expression Scaling to Enhance Emotional Communication while Mitigating Uncanny Valley Effects (RQ3)

Our selective facial expression exaggeration strategy represents a novel approach to enhancing emotional communication in social VR while mitigating uncanny valley effects. By targeting specific facial regions rather than applying uniform exaggeration, this method offers a more nuanced and effective way to improve avatar expressiveness.

The strategy's impact on emotional perception was particularly significant for negative emotions. For Fear, Disgust, and Anger, our approach substantially enhanced emotional intensity and contagion, indicating improved emotional communication. This enhancement is crucial in social VR contexts, where accurate emotion perception is vital for

fostering empathy and building meaningful interactions. By effectively amplifying typically subtle expressions, our strategy addresses a key challenge in avatar-mediated communication, potentially leading to more engaging and authentic virtual social experiences.

Regarding the uncanny valley effect, our approach proved effective in enhancing emotional expressiveness and realism for negative emotions, thereby reducing uncanny valley effects. For positive emotions like Joy and Surprise, while improvements were less pronounced, the strategy maintained a neutral or slightly positive impact without exacerbating uncanny valley effects. This balanced approach is significant as it enhances expressiveness where needed without compromising the naturalness of already expressive emotions, striking a crucial balance in avatar design.

Our focus on selective enhancement of facial areas, rather than different emotions, offers several advantages while also presenting some challenges. This regional approach provides a universal strategy that doesn't rely on potentially inaccurate emotion recognition systems, making it more robust and versatile for real-time applications in social VR. It's simpler to implement and can be applied across various emotional states without the need for complex emotion classification algorithms. However, this approach also has limitations. The varied effectiveness across emotions highlights the complexity of human expressions and the challenges in developing a one-size-fits-all solution. While effective for some emotions, it showed limited impact on others and occasionally introduced minor negative effects. This underscores the need for a more nuanced, potentially emotion-specific approach to facial expression scaling in virtual avatars, challenging the conventional uniform approach in avatar design.

These findings have broader implications for the field of social VR and avatar design. They suggest that future avatar systems might benefit from a hybrid approach, combining region-specific enhancements with emotion recognition technologies. Such a system could dynamically adjust facial expression exaggeration based on both the detected emotion and the specific facial regions most crucial for that emotion's expression. This could lead to more sophisticated and context-aware avatar expression systems, further enhancing the quality of social interactions in virtual reality.

Moreover, our research aligns with and can inform recent developments in multimodal algorithms for emotion recognition and facial expression generation [14, 79, 80, 98]. By providing insights into how different emotions activate specific facial regions and the appropriate amplitudes for movement, our findings can contribute to the development of more naturalistic and effective avatar expressions in these systems.

6 LIMITATIONS AND FUTURE WORK

Research has shown that cultural background, gender, and age can impact facial expressions and emotion recognition [19, 48]. In our study, emotional expressions were sourced from actors and participants who were exclusively Chinese university students. Future research should extend the participant pool to include a broader range of ages and ethnic backgrounds to verify our findings across a more diverse population.

Additionally, our study focused solely on cartoon avatars. Some studies indicate that emotions in realistic virtual characters are perceived as more intense than those in lower-fidelity counterparts [35], and increased avatar realism could potentially reduce the Uncanny Valley effect [8, 35]. Current immersive social applications also feature high-fidelity virtual representations, such as Persona³ on Apple's Vision Pro. Therefore, it is essential to verify our conclusions using avatars with varying levels of realism to assess their generalizability.

In terms of experimental design, we opted to use pre-recorded emotional expressions performed by actors and replayed in a simulated face-to-face virtual environment to ensure controlled variables and participant focus. Although

³<https://support.apple.com/guide/apple-vision-pro/capture-your-persona-beta-dev934d40a17/visionos>

no participants reported discrepancies between this method and their experiences in social VR communication, this setup differs from genuine two-way interactive dialogues, and our findings might not generalize to such contexts. Additionally, despite multiple rounds of script discussions and revisions to ensure emotion consistency, variations between scripts may have influenced our results. Furthermore, our research concentrated on one-on-one interactions, while emotion transmission dynamics could differ in group conversations [6]. Therefore, our study should be seen as an initial exploration, necessitating further research that examines and validates facial expression scaling techniques in naturalistic dyadic or multi-party emotional dialogues.

Finally, our study primarily examined the impact of facial expression scaling on the listener, but the self-perception of the speaker is equally crucial [26]. The effects of expression scaling on speakers remain an avenue for future research.

7 CONCLUSION

In conclusion, our study offers significant insights into the role of facial expression scaling in social VR environments, specifically focusing on cartoon avatars. Through a rigorous mixed-methods approach, we found that exaggerated facial expressions yield mixed results: they can both increase and decrease the uncanny valley effect depending on the emotion being conveyed. Our results showed that while exaggerated expressions decrease realism and comfort for emotions like Joy and Surprise, they enhance realism and appeal for emotions such as Fear, Sadness, and Anger. These findings led us to develop a selective facial expression exaggeration strategy, which effectively mitigates the uncanny valley effect for negative emotions while maintaining or slightly enhancing appeal for positive emotions. Our study provides actionable guidelines for designers and developers to improve the emotional realism and user experience of cartoon avatars in social VR, paving the way for more nuanced and expressive virtual interactions.

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